GARY CHANG
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Data and geospatial scientist applying Bayesian spatial modeling, causal inference, survival analysis, and network methods to political and policy questions where data is incomplete or deliberately concealed. Substantive focus: comparative authoritarianism, elite politics, and Indo-Pacific geopolitics.

EDUCATION

- **Ph.D., Political Science**, University of Texas at Dallas
- **M.S., Geospatial Information Science**, University of Texas at Dallas
- **M.S., Social Data Analytics and Research**, University of Texas at Dallas
- **M.S., Computer Engineering**, Ohio State University
- **M.B.A., Strategic Marketing**, San Francisco State University
- **M.S., Thermodynamics**, Shanghai Jiao Tong University
- **B.S., Marine Power Engineering**, Shanghai Jiao Tong University

CERTIFICATES

- Graduate Certificate in Geospatial Information Sciences, UT Dallas (2025)
- Certificate in Graduate Teaching, UT Dallas (2024)

DISSERTATION PAPERS

- **Geography of Power: Spatial Pathways to the CCP Central Committee** (2026) — Tests whether elite production across 2,800+ Chinese counties (16th–20th Party Congresses, 2002–2022) reflects spatial spillover, accessibility, or embedded institutional advantage. Neighbor-diffusion models (BESAG/BYM2) add little; proximity to national centers and 1960s railway access matter, and geographic advantage persists almost perfectly across congresses (AR1 $\rho \approx 0.997$). *Methods:* GLM, Bayesian INLA, CAR/BYM2, AR1 spatiotemporal models, LLM-assisted data construction, ArcGIS Pro historical railway digitization, Google Earth Engine.
- **Who Survives? Elite Retention and the Mechanisms of Durability in the CCP Central Committee** (2026) — Examines which Central Committee members are retained into the next congress. Age-based retirement norms still bind under Xi, while dynamic network embeddedness outpredicts static factional labels; a documented tie to Xi is the most consistently significant relational channel, and exit rises sharply at the 18th Congress. *Methods:* Logistic regression, Cox proportional hazards with frailty and stratified baselines, mixed-effects logistic models, relational embeddedness index.
- **Who Advances? Elite Selection and the Transformation of CCP Promotion Logic** (2026) — Models promotion from the Central Committee to the Politburo (281 eligible observations; extended panel 1982–2022). Communist Youth League affiliation reverses from asset to liability after the 18th Congress (≈42% vs 8% promotion probability before; ≈9% vs 17% after), while Xi's influence runs through individual ties rather than a cohesive faction. *Methods:* GLM/GLMM, cause-specific Cox competing-risks models, network density decomposition, QAP, exploratory ERGM.

APPLIED AND METHODS RESEARCH

- **Flight Dispatcher Annual Planning: Optimizing a Year of Dispatcher Bidlines with Mixed-Integer Programming** (2026) — *Team project, American Airlines OR & AA COMPASS Challenge (with Minh Dao and Sadman Shafiq; advisor Dohyeong Kim)* Formulated a lexicographic MILP that schedules a year of dispatcher bidlines across 8 work centers and 3 shifts (24 blocks) under a 4-3-4-4 work pattern, 4 flex days, a 27±2 daily vacation quota, and a 12-seat daily training limit. Flex days cut required bidlines from 494 to 455 (−7.9%) and halved surplus staffing; the full plan reached 100% coverage with vacation spread far more evenly than AA's historical schedules (sd 0.48 vs 3.0). Deployed on Slurm HPC and as a Nextmv decision app. *Methods and tools:* MILP, lexicographic multi-objective optimization, iterative decomposition, Python, FICO Xpress, CBC/PuLP, Nextmv, Snowflake
- **Mitigating Flight Sequence Risk through Spatial-Temporal Modeling and Optimization** (2026) — *Top-7 Finalist, American Airlines / UTD Challenge* Built a 7M+ connection-pair dataset from BTS records (2010–2019) for American Airlines' DFW hub and predicted misconnection risk (XGBoost AUC = 0.988). Fed risk scores into a Google OR-Tools integer program that cut average predicted connection risk by 98% and avoided 78,000+ high-risk itineraries.
- **Geospatial Sentiment Analysis of the 2022 Qatar World Cup on Twitter** (working paper, 2026) — Replicates and extends Hassan and Wang (2024) to 3.6M multilingual tweets with RoBERTa sentiment scoring and a three-source geocoding chain (GPS, place ID, Nominatim/OpenStreetMap). Maps sentiment by country and political bloc; scored on a university HPC cluster via SLURM array jobs in about 10 hours.
- **Bayesian Spatiotemporal Modeling of Renal Disease Mortality Across Texas Counties, 2010–2019** — (master's thesis paper, 2025) Compares eight Bayesian hierarchical specifications across 254 Texas counties. The preferred spatial-RW1 model raises R^2 from 0.014 to 0.421 and removes residual autocorrelation, revealing persistent excess mortality in South Texas beyond income, education, and care access.
- **System Justification in a Non-Democracy: Evidence from China** (field research paper, 2024) — Tests system justification theory with CGSS 2015 and 2021. Contrary to the theory, politically privileged groups support migration and speech restrictions while the powerless lean against them; education weakens the privilege effect.
- **Detecting Objects in Drone Photos with YOLO, Roboflow, and Label-Studio** (2025) — Developed PyYolo, a Python CLI that unifies labeling, format conversion, YOLOv8 training, and inference into a reproducible pipeline; applied to construction-site drone imagery (peak F1 = 0.78).
- **Damage Assessment of Wenchang Space Center after Typhoon Yagi** (2024) — Used pre- and post-storm Sentinel-2 imagery with NDVI change detection in Python; CLAHE dehazing lowered the estimated damaged area from 63.2% to 50.6%, showing how haze inflates apparent loss.
- **Siting Hybrid Renewable Energy Systems in Tanzania** (2023) — Built a ten-criterion weighted suitability model in ArcGIS Pro from 15+ data sources, identifying two consistent high-suitability clusters for hybrid solar-wind plants.

WORKING PAPERS

- Chang, Gary. 2025 — "The Price of Being Right: Why Moral Superiority Can't Win Global Power." SSRN Working Paper. doi:10.2139/ssrn.5667512
- Chang, Gary. 2025 — "Autocratic Survival Beyond Institutions: History, Culture, and Propaganda." SSRN Working Paper. doi:10.2139/ssrn.5523118

TEACHING

- **Instructor / Teaching Assistant, University of Texas at Dallas** (9/2020–present) — Political Economy, American National Government, State and Local Government, Research Design I (undergraduate).
- **Foreign Scholar, Peking University, School of Software and Microelectronics** (8/2013–1/2016) — Team Skills (graduate): debate, proposal writing, strategic marketing, presentation.
- **Lecturer, Northwood University (China program,** 8/2013–1/2016**)** — Financial Institutions and Simulation, Real Estate Finance & Analysis (undergraduate).

PROFESSIONAL EXPERIENCE

- **Research / Teaching Assistant**, University of Texas at Dallas, Dallas, TX (2020–present) — Applied research across political science, public health, GIS suitability analysis, remote sensing, and operations research using classical statistics, machine learning, network analysis (QAP/ERGM), and Bayesian/INLA methods.
- **Senior Software Engineer**, WheelsAmerica, Dallas, TX (8/2018–12/2021) — Built codegen tooling for data mappings and RESTful interfaces (PHP, Phalcon, AWS) linking inventory and accounting systems with eBay, Amazon, and SureDone.
- **Marketing Advisor**, VSO / UN Tanzania Local Enterprise Development (T-LED), Mwanza, Tanzania (3/2016–7/2018) — Supported SME growth across Malawi, Rwanda, Uganda, and Tanzania through market research, training, business plans, and fundraising proposals.
- **Foreign Scholar**, Peking University, School of Software and Microelectronics, Beijing (8/2013–1/2016) — Led the "Classroom Automation" project (LAMP stack) for the National Challenge Cup, covering mobile submission, auto-grading, and resource sharing.
- **Senior Director of Globalization**, American Express, New York (1/2010–3/2013) — Led global deployment of the SERVE online payment platform across legal, compliance, I18N/L10N, and IT workstreams.
- **VP of Engineering**, Xubo Media Ltd, Hong Kong (7/2006–1/2010) — Built the ABITO mobile ad platform and AdEstate cross-channel ad trading system.
- **Project Officer**, Plateau Homeland Conservation, Nyingchi (Tibet) (8/2005–7/2006) — Coordinated fundraising for livestock infrastructure projects.
- **Partner**, Mercatela Ltd, Tokyo (3/2003–8/2005) — Built Foodbiz.com for Nikkei Business Publications (J2EE, Oracle, WebSphere).
- **Principal Consultant**, IBM, Burlingame, CA (2/2000–2/2003): Defined CRM/SCM requirements and EAI architecture, build the systems for Manugistics, BetzDearborn, and US West.
- **Senior Software Engineer**, SAP Technology, Palo Alto, CA (2/1996–1/2000) — Led the Sales Force Automation design team.

AWARDS AND CERTIFICATIONS

- Fred Hill Endowed Scholarship, UT Dallas (2025)
- Top-7 Finalist, American Airlines / UTD Challenge (2026)
- Esri certificates: Deep Learning Using ArcGIS Pro, Spatial Data Science, Hot Spot and Cluster/Outlier Analysis, Random Forest and Regression Modeling in ArcGIS Pro, Weighted Suitability Modeling, R-ArcGIS Bridge

SKILLS

- **Quantitative methods:** OLS/GLS/GLM/GLMM, causal inference (DID, RDD, PSM, 2SLS, synthetic control); PCA/factor analysis, Ridge/Lasso; Bayesian spatiotemporal modeling (INLA); survival analysis (Cox, frailty, competing risks); network analysis (QAP, ERGM).
- **Optimization:** mixed-integer linear programming (MILP), lexicographic multi-objective optimization, FICO Xpress, CBC/PuLP, Google OR-Tools, Nextmv.
- **Machine learning and AI:** Supervised and Unsupervised methodologies, such as KNN, Random Forest, XGBoost, SVM, Bagging, K-Means, etc.; Deep Learning (CNNs) for image classification and recognitions; transformer NLP (RoBERTa), LLM API pipelines for data extraction; agentic workflows (LangGraph, Cursor, Claude Code); YOLO object detection.
- **Geospatial:** ArcGIS Pro, QGIS; Google Earth Engine, remote sensing and change detection, Drone2Map; suitability modeling/MCDM; Spatial and Areal Interpolation; Image Enhancement and Processing.
- **Programming and data:** R, Python, Stata; SQL (MySQL, PostgreSQL), Snowflake; PHP, Java, C/C++/C#; HPC/SLURM; Tableau, Power BI.

REFERENCES

- **Thomas Gray**, Professor, School of EPPS, UT Dallas · Thomas.Gray1@utdallas.edu
- **Banks Miller**, Professor, School of EPPS, UT Dallas · millerbp@utdallas.edu
- **Clint Peinhardt**, Professor, School of EPPS, UT Dallas · clint.peinhardt@utdallas.edu
- **Michael Tiefelsdorf**, Professor, Geospatial Information Sciences, UT Dallas, tiefelsdorf@utdallas.edu
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